

Assignment

Vehicle Rental Price Calculation System

This assignment aims to deepen your understanding of microservices by applying them to a practical scenario involving dynamic data manipulation and service interaction.

Overview

In this project, we will continue to explore the concepts of microservices that we previously engaged with, by developing a new application. This application will manage and manipulate data from a database that contains dynamic pricing information for various vehicle rentals across different seasons.

Previous Project Recap

As you remember, our last project dealt with a database storing exchange rates for different currencies. We developed microservices that interacted with this database to retrieve and display exchange rates, perform calculations based on given quantities, and present the results. This included handling specific URLs and ports for queries such as retrieving exchange rates between two currencies and calculating total amounts.

New Project Details

For this assignment, we will focus on a subsystem within a vehicle rental marketplace that manages rental prices over different seasons. You have the option to import this data from an Excel spreadsheet or a JSON array of objects. Here's what you need to know:

1. Database Setup:

- **Excel Option:**
 - You will be provided with an Excel spreadsheet containing the rental prices of approximately 50 different types of vehicles for each season of the year.
 - The first column lists the vehicle types, and the first row lists the seasons (Spring, Summer, Fall, Winter).
 - Prices are given in dollars per day for each vehicle type for each season.
- **JSON Option:**
 - Alternatively, you can use a JSON array of objects with the same data structure.
 - Each object will represent a vehicle type with properties for each season indicating the daily rental price.
- You will import this data into your database or directly use it in your application as needed.

2. Microservice Design:

- **Vehicle Season Price Microservice:** Retrieves the daily rental price of a specific vehicle for a specific season from the database or the JSON data.
- **Vehicle Total Price Microservice:** Calls the Vehicle Season Price Microservice to retrieve the price, then calculates the total rental price based on a given rental period.
- Example queries you might handle:
 - Retrieve the daily rental price of an SUV in Summer.
 - Calculate the total rental cost for renting an SUV for 10 days in Winter.

3. Requirements:

- **Data Import/Use:** Choose to import the provided Excel data into your database or use the JSON data directly in your application. A sample dataset can be something as follows:

	A	B	C	D	E
1	Vehicle Type	Spring	Summer	Fall	Winter
2	Compact	\$45	\$50	\$43	\$40
3	Sedan	\$55	\$60	\$53	\$50
4	SUV	\$65	\$70	\$63	\$60
5	Convertible	\$75	\$80	\$73	\$70
6	Truck	\$85	\$90	\$83	\$80

OR

```
[
  {
    "vehicleType": "Compact",
    "pricing": {
      "Spring": 45,
      "Summer": 50,
      "Fall": 43,
      "Winter": 40
    }
  },
  {...}, {...}
]
```

- **Microservice Implementation:**
 - Implement the 'Vehicle Season Price' microservice to fetch the daily rental price for any given vehicle and season.
 - Implement the 'Vehicle Total Price' microservice to calculate and return the total rental price for a specified period of a vehicle for a given season.
- **Standardization:** Establish and utilize a standard port configuration for your microservices similar to what we used for the currency exchange microservices.
- **Output:** Each microservice should provide outputs in JSON format, detailing the price data or calculations along with the environment settings such as the port number.
- **Presentation Requirement:** Students must also present their project, which includes explaining the code, demonstrating the application in action, and showcasing the results. This presentation can be conducted live via a Zoom session or can be submitted as a recorded video along with the source code.

Assignment Deliverables

- Complete the microservice implementations as described.
- Provide a document detailing your API endpoints, including example requests and responses.
- Submit your code along with instructions on setting up and running your microservices.
- Prepare and deliver a presentation that demonstrates the functionality of your application and explains your code.

Submission Guidelines

- Ensure your code is well-documented.
- Submit your project via the course portal by the deadline provided.
- Include any assumptions or additional features you implemented in your project documentation.
- Submit your presentation video link or schedule your live session through the course portal as part of your final submission.
- Please respect the deadline for submission as it cannot be extended.

I'm excited to see the innovative solutions you come up with. Good luck!